# How AdamW and Muon Train a Small Language Model

Next steps (ii) and (iii) of the checkpoint

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**Summary.** The Step 2 runs reproduce on a new cluster. (ii) The update cosine comes from the rule alone. Stable rank and $\sigma_1/\sigma_{\text{median}}$ follow the rule too, but only on real gradients; the spectral norm does not. (iii) Weight decay does not explain Muon's speed, and orthogonalization is necessary for it. Whether the spectra cause the speed is open.

## 1 Setup and reproduction

TinyStories-10M, the checkpoint's settings: AdamW at $2 \cdot 10^{-3}$, Muon at 0.01, 10,000 steps, data orders 0 to 2. Run on FAS RC Cannon. The rerun agrees with the checkpoint (Table 1).

**Table 1:** Rerun (checkpoint in brackets); metrics at training loss 1.81.

| | AdamW | Muon |
|---|---|---|
| final validation loss | 1.339 to 1.345 (1.344) | 1.308 to 1.310 (1.309) |
| step reaching AdamW's final loss | | 7250 to 7750 (7500) |
| stable rank of $W$ | $32.7 \pm 0.2$ (33.0) | $86.3 \pm 0.5$ (86.2) |
| active MLP units | $0.120 \pm 0.001$ (0.120) | $0.222 \pm 0.001$ (0.222) |

## 2 (ii) Direct result of Muon's updates, or emergent in training?

### 2.1 Rules without learning

Each rule updates one block matrix of each shape for 10,000 steps on synthetic gradients: noise, or a rank-4 direction plus noise. Nothing is learned (Fig. 1).

**Update cosine follows the rule.** On noise, consecutive updates have cosine 0.90 under AdamW and 0.66 to 0.76 under Muon, close to the real runs (0.89 and 0.74).

**The spectra do not.** On noise, neither rule moves the stable rank of $W$. With the rank-4 signal both fall to about its rank, AdamW's within a few hundred steps and Muon's by the end (step 1000: 3.7 against 9.9).

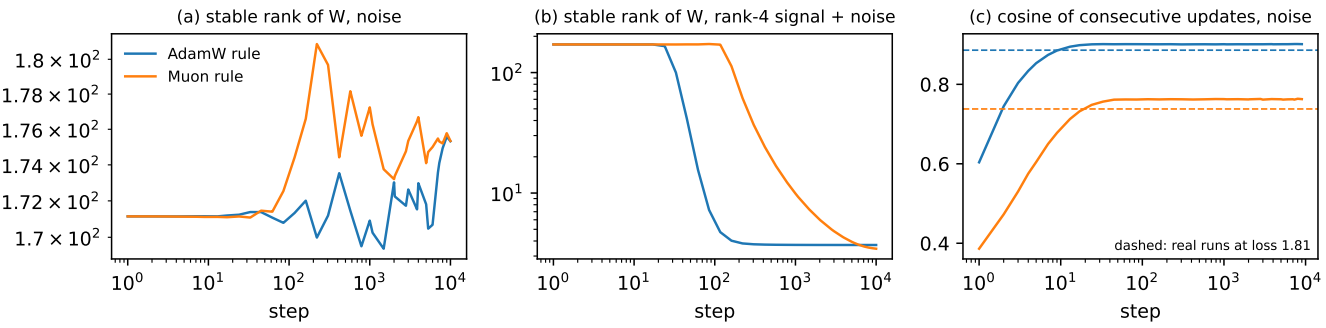



**Figure 1:** Each rule on synthetic gradients, MLP input matrix.

### 2.2 Ghosts on real gradients

Each run carries a *ghost*: a copy of the initial weights updated by the other optimizer's rule on the run's own gradients. The ghost never feeds back into the run. With the two runs this makes a $2 \times 2$ design, rule $\times$ gradient stream (Table 2, Fig. 2).

**The ordering follows the rule.** Muon's rule on AdamW's gradients ends at stable rank 87, against 40 for the AdamW run and 75 for the Muon run. AdamW's rule on Muon's gradients ends at 13. $\sigma_1/\sigma_{\text{median}}$ and update cosine go the same way. Both ghosts go further than the run with their rule (87 against 75, 13 against 40), so learning pulls the two optimizers together.

**The spectral norm does not.** On AdamW's gradients Muon's rule gives 10.9 and AdamW's 5.9; on Muon's gradients, 7.5 and 7.8. Which rule is larger depends on the gradient stream.

**Not shown.** A ghost has no loss or Hessian, so it says nothing about $\lambda_{\max}$, MLP sparsity or the loss.

**Table 2:** The $2 \times 2$ design at step 10,000.

| rule | gradient stream | stable rank | $\sigma_1/\sigma_{\text{median}}$ | update cosine |
|---|---|---|---|---|
| AdamW | AdamW (run) | 39.9 to 41.4 | 5.80 to 6.19 | 0.89 to 0.90 |
| AdamW | Muon (ghost) | 12.6 to 13.0 | 15.6 to 16.0 | 0.89 |
| Muon | Muon (run) | 75.3 to 75.5 | 2.89 to 2.90 | 0.71 |
| Muon | AdamW (ghost) | 87.0 to 87.2 | 2.77 to 2.80 | 0.70 |

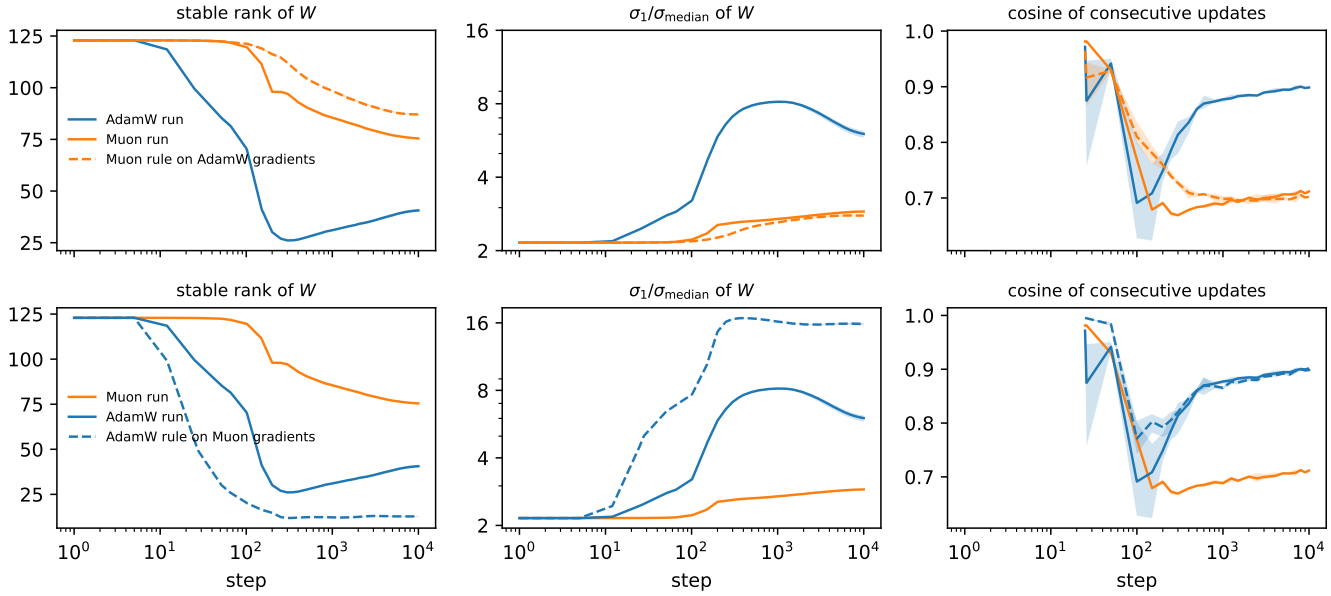



**Figure 2:** Ghosts (dashed) against runs.

# 3 (iii) Do the differences explain Muon's faster training and different path?

## 3.1 What the speed needs

Three ablations: AdamW without weight decay, Muon with weight decay 0.02, and Muon without orthogonalization (Table 3).

**Weight decay is not the reason.** Muon has none and is already faster; removing it from AdamW costs a little (1.346 to 1.353 against 1.339 to 1.345) and adding it to Muon helps a little (1.297 to 1.304 against 1.308 to 1.310).

**Without orthogonalization Muon trains worse than AdamW.** At learning rate 0.01 the final loss is 3.729, 2.943 and 1.389, and no rate tried (three, on one data order) reaches AdamW's 1.339. So orthogonalization is necessary for Muon's speed.

**Table 3:** Final validation loss at step 10,000.

| run | data order 0 | 1 | 2 |
|---|---|---|---|
| AdamW (baseline) | 1.339 | 1.341 | 1.345 |
| AdamW, weight decay 0 | 1.346 | 1.353 | 1.350 |
| Muon (baseline) | 1.308 | 1.309 | 1.310 |
| Muon, weight decay 0.02 | 1.297 | 1.298 | 1.304 |
| Muon, no orthogonalization, lr 0.01 | 3.729 | 2.943 | 1.389 |
| Muon, no orthogonalization, lr 0.003 | 1.454 | | |
| Muon, no orthogonalization, lr 0.03 | 4.078 | | |

### 3.2 Dose-response, swap timescale, loss decomposition

Three runs are in progress and have no results yet: a dose-response in Muon's singular-value power, from exact orthogonalization to none; the swap timescale, whether the loss rate changes at once after a swap as the update cosine does, or with the spectra; and a per-step loss decomposition.